Hello!

Lecture 12 - Inverse and Implicit Differentiation

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Updates

Curve information forthcoming - in contact with course coordinator about adjusting final grade distribution.
Regrade requests for Exam 1 closed.

Quiz Time!

Let's get into neat rows, spread out, and slay this quiz!

Inverse and Implicit Differentiation

The Road So Far

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Combination Rules:

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Combination Rules:

- Sum/Difference Rule: $\frac{d}{dx}(f(x) \pm g(x))$
- Product Rule: $\frac{d}{dx}f(x)g(x)$
- Chain Rule: $\frac{d}{dx}g(f(x))$
- Constant Multiple Rule: $\frac{d}{dx}c \cdot f(x)$
- Quotient Rule: $\frac{d}{dx}\frac{f(x)}{g(x)}$
- Inverse Function Rule: $\frac{d}{dx}f^{-1}(x)$

Explicit vs. Implicit Definitions I

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- $f(x) = x^2$
- $g(x) = \sin(x)$
- $h(x) = \sqrt{x^2 + 2x}$
- $j(x) = e^{\sin(x)}$
- $k(x) = (x + \sin(x))^{\frac{1}{2}}$
- etc

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This is not the only way in which we can define functions!

Explicit vs Implicit Definitions II

We could instead define relations *implicitly* - i.e. by writing out some equation that we want the function to satisfy, e.g.:

Explicit vs Implicit Definitions II

$y = f(x)$



$x^2 + y^2 = 1$

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- $x^2 + y^2 = 1$
- $y^2 = x^3 + 2x + 1$
- $\arctan(\frac{y}{x}) = \frac{\pi}{4}$
- $\sin(x) + \cos(y) = 1$
- $x^2 - y^2 = 2$
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Just as we can *graph* a function: i.e. we plot all the points $(x, f(x))$, we can plot some implicit equation $F(x, y) = 0$ by plotting all the points (x, y) so that $F(x, y) = 0$.

Implicit diff.

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When is this useful?

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$$\sum_{i=1}^{n} m_i v_i^{init} = \sum_{i=1}^{n} m_i v_i^{final} \quad (1)$$

$$\sum_{i=1}^{n} \frac{1}{2} m_i (v_i^{init})^2 = \sum_{i=1}^{n} \frac{1}{2} m_i (v_i^{final})^2 \quad (2)$$

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Also: predator-prey models, resource allocation, budgeting, etc.

Inverse Functions as a Relation

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To address our final desired synthetic rule (inverse function rule)

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  1. $f(f^{-1}(x)) = x$
  2. $f^{-1}(f(x)) = x$

Inverse Functions as a Relation

$f: D \to R$ $\quad f^{-1}: R \to D$ $\quad e^x \rightsquigarrow \ln(x)$

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  i.e. $f(x)$ and $f^{-1}(x)$ "undo" eachother.

$\rightsquigarrow$ Inverse function rule

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4. If $f(x) = e^x$ with domain $\mathbb{R}$ and range $(0, \infty)$ $f^{-1}(x) = \ln(x)$.

↳ Domain $(0, \infty)$

Range: $\mathbb{R}$

Inverse Differentiation I

Question: Using the relation

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how might we relate the derivative of $f(x)$ with the derivative of it's inverse function $f^{-1}(x)$?

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Inverse Differentiation II

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$y = g(x)$

$x = g^{-1}(y)$

$1 = g'(g^{-1})'(y)$

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Inverse Differentiation III

Ex. $\frac{d}{dx}\ln(x)$.

Derivative of Natural Logarithm

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$\ln(x)$ is inverse to $e^x \implies f(x) = e^x, f^{-1}(x) = \ln(x)$.

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Derivative of Natural Logarithm

$$\frac{d}{dx}\ln(x) = \frac{1}{x} \tag{12}$$

Inverse Differentiation IV

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$$\frac{d}{dx}\arctan(x) = \frac{1}{\sec^2(\arctan(x))} = \frac{1}{1+x^2} \tag{16}$$

memorize

derive

Inverse Differentiation V

Derivatives of Inverse Trigonometric Functions

$$\frac{d}{dx}\arcsin(x) = \frac{1}{\sqrt{1-x^2}} \tag{17}$$

$$\frac{d}{dx}\arccos(x) = -\frac{1}{\sqrt{1-x^2}} \tag{18}$$

$$\frac{d}{dx}\arctan(x) = \frac{1}{1+x^2} \tag{19}$$

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$$x^2 + y^2 = 1 \tag{20}$$

Differentiate both sides w.r.t. x (21)

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Assuming $y = y(x)$

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$$\frac{d}{dx}(x^2 + y^2) = \frac{d}{dx}1 \tag{22}$$

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = 0 \tag{23}$$

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$x \xrightarrow{y(x)} y(x) \xrightarrow{x^2} y(x)^2$

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Differentiate both sides w.r.t. x (21)

$$\frac{d}{dx}(x^2 + y^2) = \frac{d}{dx}1 \quad 0 \tag{22}$$

$2x$

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = 0 \tag{23}$$

$$2x + \frac{d}{dx}(y(x)^2) = 0 \tag{24}$$

Implicit Differentiation I

What about for the other kinds of implicit relations?

Q: At a point (x, y) on the circle of radius 1, what is the slope of the line tangent to the circle at that point?

Circle of radius 1 $\implies x^2 + y^2 = 1$.

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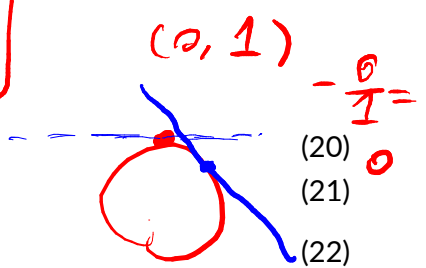
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The core idea is that in an implicit relation $F(x, y) = 0$, we can *always* treat one of the variables as a function of the other (fancy reason: *implicit function theorem*) and thus can differentiate implicit relations just by using the chain rule!
We'll see more application oriented examples when we do related rates soon :)

Worksheet Time

Let’s get into groups and work on the worksheet!

End of Class

That’s all folks!